

Saving Water Indoors

by Peter Mayer

Got water conservation? If not, you most likely will in the coming decade. According to a national survey conducted by the U.S. Government Accountability Office, 36 states anticipate some sort of water shortage by 2013. The latest climate projections from the National Oceanic and Atmospheric Administration show intensifying drought scenarios for large parts of North America in the coming decades. We must learn to conserve water now if we are to sustain our way of life and preserve our environment for future generations.

Efficient use of water is particularly important in our homes, as many fairly easy opportunities exist there for conservation. In most U.S. cities, single-family homes make up the single largest source of demand for water. Many homes built before 1994 are equipped with inefficient plumbing fixtures and appliances that use 30%–40% more water than modern, efficient units. A recent field study, which was funded by the federal EPA's Office of Water, found that many people can substantially reduce their water use without altering their behavior or lifestyle simply by installing efficient fixtures and appliances. Payback time for these efficiency upgrades is typically less than five years.

The company I work for, Aquacraft, Incorporated, conducted the residential water efficiency study mentioned above. Aquacraft is an engineering firm that specializes in measuring water use patterns and evaluating the effect of water conservation programs. We examined indoor water use in 96 single-family homes in Seattle, Oakland, and Tampa before and after installing water-conserving toilets, clothes washers, showerheads, and faucet aerators. The participating



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homes had water use patterns that were statistically similar to those of single-family customers in each of the participating water utilities' territories. The participating utilities were the East Bay Municipal Utility District (EBMUD), in Oakland, California; the Tampa Water Department, in Tampa, Florida; and Seattle Public Utilities, in Seattle, Washington.

Aquacraft has developed a water use measurement technique known as flow trace analysis. This method measures the water used by each fixture and appliance in the home by continuously recording flow through the water meter, using a battery-powered data logger, and then breaking down the resulting flow trace data stream, using specially designed signal processing software. The result is a measurement of the volume of water used each time somebody flushes a toilet, runs a clothes washer, or takes a shower. This

technique has been used successfully in many research studies, including the often referenced *Residential End Uses of Water Study* (REUWS) published by the American Water Works Association in 1999.

SETTING UP THE SURVEY

Between 30 and 35 homes were selected to participate in each of the three cities; the same basic research methods were used in each city. Invitations to participate in the project were sent to a sample of customers. Participants were offered the chance to keep the new water-efficient fixtures and appliances that we would be installing and monitoring as an inducement to cooperate. The costs of buying and installing these efficient new fixtures and appliances were borne by the participating water utilities (a

value of approximately \$1,500 per home).

A project team then visited each home to obtain baseline water use data and to conduct the household survey and get responses to a series of roughly 40 auditing questions. These questions solicited information about the size and composition of the household, the existing fixtures and appliances, and any habits or hobbies of the residents that might affect water use. This questionnaire also asked the residents whether they were satisfied with their existing toilets, clothes washers, and other fixtures before these were replaced with new models.

Historic water billing data were obtained for each home to get a sense of how much the residents' typical monthly bills had been. In addition, 15 days of detailed end use data were obtained for each home before any new fixtures were installed. In 20 of the 96 homes, a special meter was placed on the tank of the water heater to measure hot water use before and after installation of the new equipment. This enabled us to measure the amount of hot water that was used for different purposes at the same time as we monitored total water use at the main meter. These data helped us to calculate the energy savings that might be achieved by using less hot water.

LET THE RETROFITS BEGIN

After we completed the initial surveys and collected the baseline data, plumbing contractors hired by the project team began to replace the fixtures and appliances. This key step required the most coordination with, and cooperation from, the residents. Many of the fixtures and appliances used in the study were purchased by the participating water utilities, using the best pricing they could obtain on the retail market. The exception to this was in Tampa, where manufacturers

donated most of this equipment, and they did so with no strings attached, leaving the study team free to conduct research independently, without review or input from the manufacturers. (The Whirlpool Corporation donated 26 clothes washers—Duet and Calypso models—and the Delta valve company donated 10 electronic faucets.)

We classified the fixtures and appliances by type and efficiency level; we did not review specific makes or models (see Table 1). Most of the



The top loading Whirlpool Calypso clothes washer achieved water and energy savings even though it used more water than some horizontal axis machines tested in the study.

fixtures and appliances used for the retrofits were similar in terms of rated water use performance. Tampa was the exception. There, we installed lower-flow-rated faucets and showers, and some of the clothes washers used only 18 gallons of water per load—the lowest water demand for clothes washers in the study.

NEW APPLIANCES PREFERRED

After all of the fixtures and appliances were installed, we gave the residents a few weeks to get used to their new equipment. We collected postretrofit end use data twice—the first time a few weeks after the retrofits were completed, and the second time approximately six months later. The purpose of the second data collection was to look for any changes in performance or water use patterns over time that might affect long-term water

savings. However, water use remained essentially unchanged over the six-month period.

After collecting the final data, we distributed a postretrofit product satisfaction survey to all participants. This gave them an opportunity to comment on their new equipment. Almost without exception, the participants preferred the new fixtures and appliances to their old ones, and this included the new ultralow-flow (ULF) and dual-flush toilets. (Some poorly performing low-flow toilets have given water-conserving toilets a bad reputation, but participants appeared to be happy with the newer redesigned toilets.) Not a single participant asked to have any old equipment back, which they had the right to do under the research contract.

RETROFITS SUCCESSFUL

The water savings achieved by installing water-conserving fixtures and appliances were substantial in each of the three study cities. Overall the 96 study homes reduced their indoor demand by 39% (see Figure 1). During the baseline period, the study homes used an average of 175 gallons per day (gpd) for indoor purposes, but after the retrofit, usage was reduced to 107 gpd. In Tampa, where lower-flow and lower-volume equipment was installed, water use was reduced by 46% from the baseline.

Hot Water Use

Overall hot water use was reduced by 21% as a result of the retrofit. Before the retrofit, only 5% of households used less than 40 gpd of hot water; following the retrofit, this number increased to 50%. A rather surprising finding was that the percentage of households that used a great deal of hot water did not decrease substantially. Before the retrofit, 35% of the households used 70 gpd of hot

water or more; after the retrofit, the percentage only decreased to 30%. Apparently residents who used a lot of discretionary hot water—who took long hot showers, for example, and frequent baths—did not change their behavior as a result of this study.

Toilet Water Use

The retrofits significantly reduced water used for toilet flushing. When we started, few of the homes in the study had any ULF toilets; after the retrofit, all of the toilets were ULF or dual-flush, so these results show the impact of an almost complete retrofit of the sample with respect to toilets. Household use for toilet flushing was reduced by 52%–59%, with an average reduction of 55%. In terms of gallons per household, the reductions ranged from 22 to 27 gpd, with an average of 25 gpd, or approximately 9,000 gallons per year per household. The participants went from using over 16,000 gallons of water per year to using 7,000 gallons per year for toilet flushing, as a result of the retrofit program.

One of the tired criticisms of ULF toilets has been that they don't save water because they must be flushed twice or more to do the job. This critique is simply not born out by the data. Toilet flushing frequency remained essentially unchanged from the baseline to the postretrofit period, going from 5.11 to 5.36 flushes per person per day. The slight increase in flushing frequency was not statistically significant—and not nearly great enough to negate the substantial water savings achieved by the retrofit.

Customers were substantially more satisfied with the performance of their new toilets than they were with the performance of their old inefficient commodes (see Table 2). The new fixtures scored higher in every category.

Clothes Washer Water Use

Water use for clothes washing was reduced by an average of 38% by the use of the new high-efficiency machines. The water savings realized ranged from 11.5 gpd (4,197 gallons

per year) to 15.1 gpd (5,511 gallons per year). The site with the highest savings was Tampa, where the machines with the lowest per load water use were installed. One of these was a horizontal-axis washer that used as little as 14 gallons for a standard-sized load. Before the retrofit, the clothes washers in the participating homes used an average of 35.9 gallons per load of clothes. After the retrofit, this

figure dropped to an average of 20.5 gallons per load.

It has been suggested that the savings from water-conserving clothes washers may be partly negated by an increase in the number of loads, because water-conserving washers have a smaller capacity. In this study the number of loads of laundry per person per day dropped from 0.38 during the baseline period to 0.36

Table 1. Fixtures and Appliances Used for Retrofits

Fixtures and Appliances	Type of Device	Flow Rate	Volume
Seattle, Washington			
Showerheads	Low-flow head	2.5 gpm	1.6 gpf
Bathroom faucets	Aerators	1.5 gpm	
Kitchen faucets	Aerators	2.5 gpm	
Toilets	ULF gravity		0.8/1.1 gpf
Toilets	ULF Dual flush		24 gal/load
Clothes washers	Horizontal-axis		27 gal/load
Clothes washers	Vertical-axis		
Oakland, California			
Showerheads	Low-flow head	2.5 gpm	1.6 gpf
Bathroom faucets	Aerators	1.5 gpm	
Kitchen faucets	Aerators	2.2 gpm	
Toilets	ULF flapperless		1.6 gpf
Toilets	ULF Pressure assist		22 gal/load
Clothes washers	Horizontal-axis		29 gal/load
Clothes washers	Vertical-axis		
Tampa, Florida			
Showerheads	Low-flow head	1.7 gpm with shutoff switches	1.6 gpf
Bathroom faucets	Aerators	1.0 gpm	
Bathroom faucets	Electronic valves	1.0 gpm	
Kitchen faucets	Aerators	1.5 gpm	18 gal/load
Toilets	ULF flapperless		27 gal/load
Clothes washers	Horizontal-axis		
Clothes washers	Vertical-axis		

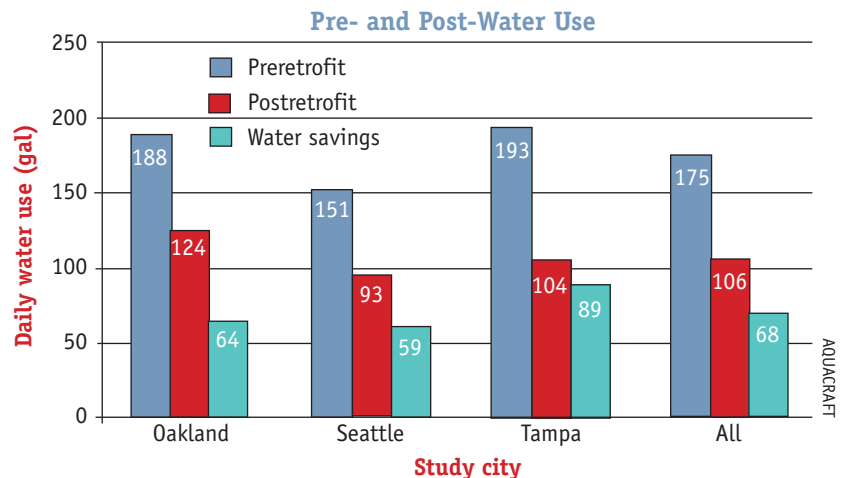


Figure 1. The water savings achieved by installing water-conserving fixtures and appliances were substantial in each of the three study cities.

Table 2. Toilet Satisfaction Ratings (Combined Group Study)

Rating Category	Preretrofit (n = 96)	Postretrofit (n = 94)
Bowl cleaning	3.4	4.0
Flushing performance	3.5	4.3
Appearance	3.4	4.7
Noise	3.3	4.6
Leakage	3.8	4.7
Maintenance	3.7	4.6
Overall average	3.5	4.5

Note: Rating scale from 1 to 5, where 1 = unsatisfied and 5 = completely satisfied

Table 3. Clothes Washer Satisfaction Ratings (Combined Group Study)

Rating Category	Preretrofit (n = 96)	Postretrofit (n = 94)
Cleaning of clothes	4.0	4.8
Reliability	4.3	4.7
Noise	3.2	4.6
Moisture content of clothes	3.6	4.7
Cycle selection	4.1	4.7
Capacity	4.2	4.6
Wash cycle time	NA	4.5
Detergent use	NA	4.6
Overall average	3.9	4.6

Note: Rating scale from 1 to 5, where 1 = unsatisfied and 5 = completely satisfied

after the retrofit. This suggests that the capacity of the water-conserving clothes washers was adequate and did not result in increased use.

Of all fixtures, clothes washers saved the most water, on a percentage basis, following the retrofit. In both cities where hot water use was monitored (Seattle and Oakland), hot water use from clothes washing was reduced by an average of 56%.

Although the water savings achieved by retrofitting clothes washers was significant, many customers were reluctant to spend what it would cost to purchase a more efficient model. Study participants were asked if they would be willing to pay a premium of \$150 to get an equivalent-quality, conserving washer. Twenty four percent said they would not be willing to do so, and 13% were unsure. However, when asked if they liked their new clothes washer better than their old one, 84% said yes, and 91% said that they would



The front loading Whirlpool Duet clothes washer resulted in significant water and energy savings.

recommend the new washer to a friend.

Participants were asked to rate the performance of their new machines in eight categories, ranging from cleaning ability to noise level. We compared their responses to their responses from the earlier survey where we had asked them the same questions about their old clothes washers (see Table 3). The new clothes washers rated better than the old ones in all categories in which data were available.

Shower Water Use

The data for water savings associated with showers is a lot less clear-cut than the data for the previous categories, but it does seem possible to save water by replacing the showerhead. The key is to understand the different types of showerhead that were used in each city. In both Seattle and Oakland, we used 2.5 gpm showerheads as replacements, but in Tampa we used either 1.75 gpm showerheads or hand-held 2.5 gpm showerheads with shut-off buttons. The only statistically significant water savings were realized in Tampa, where the retrofit reduced water used for showering by 28%, compared to 9% reductions in both Oakland and Seattle. The insignificant reduction in these

two cities is attributable to the fact that 2.5 gpm showerheads were used as replacements. Many participants in these cities were already showering at or below the 2.5 gpm flow rate, so the switch to the new equipment made no difference.

The results for showerheads strongly suggest that installing 2.5 gpm showerheads will not lead to significant savings in the typical single-family home, but that savings can be achieved by installing 1.75 gpm or hand-held showerheads with shutoff valves. Overall, 73% of participants said they would recommend their new showerheads to a friend, and 73% liked their new showerhead better than their old one.

The change in hot water use for showering was not statistically significant for either of the hot water monitoring sites (Seattle and Oakland). This result makes sense, since neither site showed a statistically significant reduction in total water use for showering—and most of the water used for showering is hot water.

Faucet Water Use

Water savings from faucet aerators were similar to, if more modest than, water savings from showerheads. In

Oakland, faucet aerators saved almost no water. In Seattle, faucet aerators saved 3.3 gpd (1,200 gallons per year), which was statistically significant. In Tampa, where the most aggressive faucet retrofits were done, savings were 7.0 gpd (2,550 gallons per year), or 28% of the preretrofit use. These savings were also statistically significant. Overall savings were 3.4 gpd, but these data, like those for the showerheads, reflect such a wide range of results in the three cities that it is better to look at each city separately.

Table 4. Anticipated Water Savings from Typical Retrofits

Fixture	Total Water Savings (kgal)	Hot Water Savings (kgal)
Toilets/leaks/other	21.1	1.1
Clothes washers	5.6	1.4
Shower heads	1.6	0
Faucet aerators	1.4	1.4
Total	29.7	3.9

Faucet use accounted for the greatest percentage of daily hot water use both pre- and postretrofit. In Oakland, faucet hot water use was reduced by 33% or 6.3 gpd. Annually this would yield a reduction of 2,400 gallons per household. However, cold water faucet use in Oakland increased, so it is unclear how reliable these hot water savings would be over time.

EXPECTED ANNUAL WATER SAVINGS FROM RETROFITS

From the perspective of the utility or the community, it is useful to consider the annual average savings that can be anticipated in a population of typical homes when standard fixtures and appliances are replaced with high-efficiency models (see Table 4). On average, retrofits should save just under 30 kgal (thousand gallons) per year of water per household. The bulk of these savings comes from replacing toilets and repairing leaks, followed by the replacement of clothes washers,

showerheads, and faucet aerators. Toilets and leaks account for 71% of the total water savings, clothes washers account for 19% and showerheads and faucets account for 5% each.

PAYBACK TIME

In general, it takes a total of about eight indoor water-conserving fixtures or appliances to bring a standard home up to grade as a water-conserving home. By installing these, the resident can expect to save approximately

designed to use less water and energy. These savings come without sacrificing performance or requiring changes in the residents' lifestyle. The national implications of these findings are significant. Many homes in America remain equipped with old, inefficient fixtures and appliances. If these are replaced with new, water-conserving equipment, daily water use in the average single-family home can be reduced by 69 gallons per day. This reduction translates to a national savings of 1.7 million gpd, or 5.3 million acre

Table 5. Summary of Household Retrofits and Annual Savings

Fixture	Costs			\$ Savings			Water Savings (kgal)
	No.	Unit Cost	Total Cost	Water (\$)	Energy (\$)	Water + Energy (\$)	
Toilets	2	\$363	\$ 72	\$130		\$130	21.13
Clothes washers	1	\$818	\$ 818	\$ 81	\$42	\$123	5.60
Showerheads	2	\$ 12	\$ 25	\$ 10		\$ 10	1.60
Faucet aerators	3	\$ 5	\$ 15	\$ 10		\$ 10	1.40
Total	8		\$1,584	\$230	\$42	\$272	29.73

30 kgal of water per year, for an average cash savings of \$272—\$230 per year in water and \$42 per year in energy (depending upon local water, wastewater, and energy rates). Assuming that the customer must bear the total out-of-pocket expense of this retrofit, and ignoring any discounting, rebates, or remaining economic life of the existing fixtures, the total cost of the retrofit will be about \$1,580. This represents six-year payback, assuming no increase in water or energy prices, which would shorten the payback period (see Table 5). If rebates are available or the old fixtures and appliances need to be replaced anyway, the cost of the retrofit may be close to the incremental costs, which amount to around \$500, with a payback period in the two-year range.

TAKING IT NATIONAL

Significant, cost-effective indoor water savings can be achieved by replacing old inefficient fixtures and appliances with new products that are

feet, annually. Comparable reductions can also be achieved by retrofitting mobile homes and multifamily buildings. Even homes built after 1994 can be made to save more water by installing water-conserving clothes washers. These savings decrease the demand on utilities for both drinking water supply and wastewater treatment. Fully realized, they could provide indoor water for many additional single-family households each year, without the need to pump, treat, or store additional water.

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For more information:

A copy of the complete research report for this EPA-funded residential water efficiency study can be downloaded for free from Aquacraft at www.aquacraft.com.